

Chahat Raj

PhD Candidate in Computer Science | George Mason University

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Research Interests

My research focuses on NLP, particularly the evaluation and alignment of large language models. I study socio-cultural biases and other harms in multilingual and multimodal systems and develop methods to mitigate them, advancing trust-worthy and responsible AI. My work has been published at ACL, EMNLP, AIES, AAAI, and other venues.

Education

May 2027 Aug 2022	George Mason University Ph.D. in Computer Science Advisors: Ziwei Zhu , Antonios Anastasopoulos	Fairfax, VA
Dec 2025 Aug 2022	George Mason University Masters in Computer Science	Fairfax, VA
Aug 2021 Aug 2019	Delhi Technological University Masters in Information Systems (Thesis) Advisor: Priyanka Meel	Delhi, India
May 2019 Aug 2015	Indira Gandhi Delhi Technical University for Women Bachelors in Computer Science & Engineering	Delhi, India

Experience

Aug 2026 May 2026	Amazon Applied Scientist Intern Mentor: Ren Pang , Manager: Zheng Qi > <i>A parameter-efficient safety steering technique for bedrock guardrails foundation models.</i>	Seattle, WA
Aug 2025 May 2024	George Mason University, University of Washington Graduate Research Assistant Advisors: Ziwei Zhu , Antonios Anastasopoulos , Aylin Caliskan > <i>Societal simulations, model unlearning, and cognitive alignment for bias mitigation.</i>	Fairfax, VA
Sep 2024 May 2024	University of Washington Tech Policy Lab, The Information School Visiting Scholar Advisor: Aylin Caliskan > <i>Bias, cognition, and mitigation in vision-language models.</i>	Seattle, WA
Aug 2023 May 2023	George Mason University School of Computer Science Graduate Research Assistant Advisor: Ziwei Zhu > <i>Fairness, interpretability, and robustness in transformer-based fake news detection.</i>	Fairfax, VA
Jul 2022 Jun 2021	University of Technology Sydney School of Computer Science Visiting Scholar Advisor: Mukesh Prasad > <i>Cyberbullying, AI for Aboriginals, modeling disaster fundraising, and ML for clinical diagnosis.</i>	Sydney, AU

Selected Publications

[full list here](#)

- [C.12] **VIGNETTE: Socially Grounded Bias Evaluation for Vision-Language Models** [PDF | Code] **SAC Highlight Award**
Chahat Raj, Bowen Wei, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Proceedings of the Association for Computational Linguistics: ACL '26, San Diego, California, USA [ACL Main '26]
- [C.11] **Talent or Luck? Evaluating Attribution Bias in Large Language Models** [PDF | Code]
Chahat Raj, Mahika Banerjee, Jinhao Pan, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Findings of the Association for Computational Linguistics: ACL '26, San Diego, California, USA [ACL Findings '26]
- [C.10] **Knowing Bias, Doing Better: Mitigating Social Bias in LLMs via Know-Bias Neuron Enhancement** [PDF | Code]
Jinhao, **Chahat Raj**, Anjishnu Mukherjee, Sina Mansouri, Bowen Wei, Shloka Yada, Ziwei Zhu
International Conference in Machine Learning: ICML '26, Seoul, South Korea [ICML '26]
- [C.9] **Measuring South Asian Biases in Large Language Models** [PDF | Code]
Mamnuya Rinki, **Chahat Raj**, Anjishnu Mukherjee, Ziwei Zhu
TrustNLP at ACL '26 [TrustNLP '26]
- [C.8] **Discovering Bias Associations through Open-Ended LLM Generations** [PDF | Code]
Jinhao Pan, **Chahat Raj**, Ziwei Zhu
In Proceedings of the AAAI Conference on Artificial Intelligence [AAAI '26]

- [C.7] **What's Not Said Still Hurts: A Description-Based Evaluation Framework for Measuring Social Bias in LLMs**
Jinhao Pan, [Chahat Raj](#), Ziyu Yao, Ziwei Zhu [[PDF](#) | [Code](#)]
In Findings of the Association for Computational Linguistics: EMNLP '25, Suzhou, China [EMNLP Findings '25]
- [C.6] **Toward Inclusive Language Models: Sparsity-Driven Calibration for Systematic and Interpretable Mitigation of Social Biases in LLMs** [[PDF](#) | [Code](#)]
Prommy Sultana Hossain, [Chahat Raj](#), Ziwei Zhu, Jessica Lin, Emanuela Marasco
In Findings of the Association for Computational Linguistics: EMNLP '25, Suzhou, China [EMNLP Findings '25]
- [C.5] **BiasDora: Exploring Hidden Biased Associations in Vision-Language Models** [[PDF](#) | [Code](#)]
[Chahat Raj](#), Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Findings of the Association for Computational Linguistics: EMNLP '24, Miami, Florida, USA [EMNLP Findings '24]
- [C.4] **Breaking Bias, Building Bridges: Evaluation and Mitigation of Social Biases in LLMs via Contact Hypothesis**
[Chahat Raj](#), Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu [[PDF](#) | [Code](#)]
Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society [AIIES '24]
- [C.3] **Global Voices, Local Biases: Socio-cultural Prejudices across Languages** [[PDF](#) | [Code](#)]
[Chahat Raj](#)*, Anjishnu Mukherjee*, Ziwei Zhu, Antonios Anastasopoulos
In Proceedings of EMNLP '23, Singapore [EMNLP Main '23]
- [C.2] **SALSA: Salience-Based Switching Attack for Adversarial Perturbations in Fake News Detection Models**
[Chahat Raj](#)*, Anjishnu Mukherjee*, Hemant Purohit, Antonios Anastasopoulos, Ziwei Zhu [[PDF](#) | [Code](#)]
European Conference on Information Retrieval [ECIR '24]
- [C.1] **True and Fair: Robust and Unbiased Fake News Detection via Interpretable Machine Learning** [[PDF](#) | [Code](#)]
[Chahat Raj](#), Anjishnu Mukherjee, Ziwei Zhu
Proceedings of the AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society [AIIES '23]

Under Review

- [S.2] **Debias It Yourself: Teaching LLMs Cognitive Bias-Mitigation Interventions**
[Chahat Raj](#), Sina Mansouri, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
[In Submission]
- [S.1] **Inoculation Prompting: Implicit Bias Mitigation in Large Language Models**
Yongxu Sun, [Chahat Raj](#), Aylin Caliskan
[In Submission]

Selected Projects

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| > AI, Power & Society | Auditing Diffusion Models for Generational and Representational Errors | [Report] |
| > Computer Vision | Cultural Image Translation for Human Figures | [Report] |
| > AI: Ethics, Policy & Society | Implicit Association Test in Multimodal Generative Models | [Report] |

Teaching Assistantship

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| > CS 484 Theory/Application Data Mining, GMU | Instr.: Keren Zhou | Spring '26 |
| > CS 584 Theory/Application Data Mining, GMU | Instr.: Ziwei Zhu | Fall '25 |
| > CS 108 Introduction to Computer Programming, GMU | Instr.: Sara Lynn Gopalakrishna, Bella Yin | Spring '24 |
| > CS 478 Natural Language Processing, GMU | Instr.: Antonios Anastasopoulos | Fall '23 |
| > COMP 502 Mathematical Foundations of Computing, GMU | Instr.: William D Ellis | Fall '23 |
| > CS 112 Introduction to Python Programming, GMU | Instr.: Shahnaz Kamberi | Fall '22, Spring '23 |

Student Mentorship

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| > Daksha Rajagopal [TJHSST → Purdue CS] | Modeling normative ethics and cross-cultural moral reasoning |
| > Angela Liu [TJHSST → Bryn Mawr College] | Assessing moral attributional biases in ambiguous contexts in LLMs |
| > Mahika Banerjee [TJHSST → KU Leuven] | Measuring attribution bias, coauthor on the ACL Findings '26 paper |
| > Mamnuya Rinki [MS Thesis, GMU → IntraFi] | Mitigating South Asian Biases, coauthor on TrustNLP @ ACL '26 paper |
| > Yongxu Sun [MS, UW → PhD, UW] | Inoculation Prompting by in-context learning to debias LLM decisions |
| > Diwita Banerjee [MS, GMU → DoorDash] | Assessing cuisine biases; scientific figure hallucination detection |
| > Sina Mansouri [MS, GMU] | Long Context Memory Information Retrieval in LLM Agents |
| > Sharanya Garika [CS 678 Advanced NLP] | Identifying intersectional bias in LLMs across Indo-Aryan languages |
| > Aksh Patel [CS 678 Advanced NLP] | Identifying intersectional bias in LLMs across Indo-Aryan languages |
| > Samruddhi Deshmukh [CS 795 AI Ethics] | Auditing Text-to-Image Generation Models for Social Representation |
| > Abhi Krishna [CS 795 AI Ethics] | Auditing Text-to-Image Generation Models for Social Representation |

Awards & Grants

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| <ul style="list-style-type: none">> SAC Highlight Award at ACL, '26> Outstanding Academic Award at GMU, '26> Graduate Teaching Award at GMU, '26> Outstanding Reviewer Award at EACL '26> CAHMP Fellowship at GMU, '24> Best Paper Award at MASC-SLL '24> Summer Graduate Research Award at GMU, '23> OpenAI Researcher Access Program Award '25> Travel Grant by AAAI/ACM AIES '24 & '23> Research Excellence Award at DTU '23 & '22> Graduate Scholarship at DTU '19 | <p>Vignette selected as top 15% of accepted papers.</p> <p>Received an Outstanding Academic Achievement Award.</p> <p>Received an Outstanding Graduate Teaching Assistant Award.</p> <p>Received an Outstanding Reviewer Award at EACL '26.</p> <p>Summer research on social biases in question answering.</p> <p>A Psychological View to Social Bias in LLMs.</p> <p>Explainability and Robustness of Debiasing.</p> <p>Funded USD 1,000 worth of OpenAI API credits.</p> <p>Funded USD 2,000 to attend AIES in San Jose and Montreal.</p> <p>Received three Commendable Research Awards with INR 150,000.</p> <p>Received INR 297,000 by AICTE for qualifying GATE.</p> |
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Talks

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| <ul style="list-style-type: none">> VIGNETTE> LLM Ethics: Bias and Mitigation> LLM Ethics: Bias and Mitigation> Breaking Bias, Building Bridges> Breaking Bias, Building Bridges> Breaking Bias, Building Bridges> True and Fair> Cyberbully Monitoring | <p><i>Talk at ACL '26</i></p> <p><i>Invited Lecture, CS 678 Advanced NLP, Fall '24</i></p> <p><i>Invited Lecture, CS 782 Advanced ML, Spring '24</i></p> <p><i>Talk at AAAI/ACM AIES '24</i></p> <p><i>Talk at SouthNLP '24</i></p> <p><i>Talk at MASC-SLL '24</i></p> <p><i>Talk at AAAI/ACM AIES '23</i></p> <p><i>ML4AU Research Showcase '21</i></p> | <p>San Diego, CA</p> <p>George Mason, VA</p> <p>George Mason, VA</p> <p>San Jose, CA</p> <p>Emory, GA</p> <p>Johns Hopkins, MD</p> <p>Montreal, QC</p> <p>Remote</p> |
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Program Committee

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| <ul style="list-style-type: none">> Reviewer | <p>NeurIPS '26, ARR '23-'26, CHI '26, AAAI '26, IASEAI '26, COLM '25-'26, TrustNLP '24-'26, LTEDI @ EACL '24, AIES '23-'26, CSCW '24, NeurIPS '23, ACM TIST</p> |
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